



## Rossmoyne Senior High School

Semester One Examination, 2022

Question/Answer booklet

### MATHEMATICS APPLICATIONS UNIT 1

#### Section Two: Calculator-assumed

# SOLUTIONS

WA student number: In figures

|  |  |  |  |  |  |  |  |
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In words

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Your name

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#### Time allowed for this section

Reading time before commencing work: ten minutes

Working time: one hundred minutes

Number of additional  
answer booklets used  
(if applicable):

|  |
|--|
|  |
|--|

#### Materials required/recommended for this section

##### *To be provided by the supervisor*

This Question/Answer booklet

Formula sheet (retained from Section One)

##### *To be provided by the candidate*

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators, which can include scientific, graphic and Computer Algebra System (CAS) calculators, are permitted in this ATAR course examination

#### Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

## Structure of this paper

| Section                         | Number of questions available | Number of questions to be answered | Working time (minutes) | Marks available | Percentage of examination |
|---------------------------------|-------------------------------|------------------------------------|------------------------|-----------------|---------------------------|
| Section One: Calculator-free    | 7                             | 7                                  | 50                     | 52              | 35                        |
| Section Two: Calculator-assumed | 12                            | 12                                 | 100                    | 98              | 65                        |
| <b>Total</b>                    |                               |                                    |                        |                 | 100                       |

## Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (98 Marks)

This section has **twelve** questions. Answer **all** questions. Write your answers in the spaces provided.

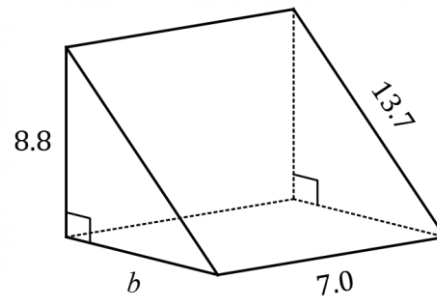
Working time: 100 minutes.

Question 8

(11 marks)

The diagram, not to scale, shows a right triangular prism with a length of 7.0 cm.

The lengths of some of the edges are shown on the diagram in centimetres.



- (a) Determine  $b$ , the length of the base of the right triangle.

(2 marks)

| Solution                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------|
| $b^2 = 13.7^2 - 8.8^2$ $= 110.25$ $b = 10.5 \text{ cm}$                                                                |
| Specific behaviours                                                                                                    |
| <ul style="list-style-type: none"> <li>✓ shows correct use of Pythagoras' Theorem</li> <li>✓ correct height</li> </ul> |

- (b) Determine the total surface area of the prism.

(4 marks)

| Solution                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Area of triangular end:</p> $A_1 = \frac{1}{2} \times 10.5 \times 8.8$ $= 46.2 \text{ cm}^2$ <p>Area of rectangles:</p> $A_2 = 7.0 \times (8.8 + 10.5 + 13.7)$ $= 7 \times 33$ $= 231 \text{ cm}^2$ <p>Total surface area:</p> $A = 2 \times 46.2 + 231$ $= 323.4 \text{ cm}^2$ |
| Specific behaviours                                                                                                                                                                                                                                                                |
| <ul style="list-style-type: none"> <li>✓ calculates area of one triangle</li> <li>✓ calculates area of at least one rectangle</li> <li>✓ indicates sum of all five faces</li> <li>✓ correct total surface area</li> </ul>                                                          |

- c) A carpenter wants to make this triangular prism out of solid wood and use jarrah, a local hardwood,  
The density of jarrah is  $835 \text{ kg/m}^3$ . (5 marks)

Calculate the weight of the triangular prism when made out of Jarrah.  
Give your answer rounded to the nearest gram. (5 marks)

| Solution                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Vol = \frac{1}{2} bh \times \text{perpendicular height}$<br>$Vol = 323.4 \text{ cm}^3$<br>$835 \div 1000000 = 0.000835 \text{ Kg/cm}^3$<br>$0.000835 \times 323.4 = 0.27 \text{ kg}$<br>$0.27 \text{ kg} \times 1000 = 270 \text{ g}$                                                                         |
| Specific behaviours                                                                                                                                                                                                                                                                                            |
| <ul style="list-style-type: none"> <li>✓ uses correct equation</li> <li>✓ Finds correct volume</li> <li>✓ Converts <math>\text{m}^3</math> to <math>\text{cm}^3</math> (<math>\div</math> by 1 000 000)</li> <li>✓ Multiplies by volume to find weight f/t</li> <li>✓ Gives correct answer in grams</li> </ul> |

**Question 9****(3 marks)**

A 2.5Kg packet of dog biscuits were on sale for \$11.20 from a veterinary shop.  
A 1.5kg packet of the same dog biscuits cost \$6.25 from a supermarket.

- a) Which is the better buy, and by how much.

| Solution                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Vet Cost per kg $\frac{11.20}{2.5} = 4.48$<br><br>Supermarket cost per kg $\frac{16.25}{1.5} = 4.17$<br>(accept any number that rounds to 4.17)<br><br>Supermarket is cheaper by 31 cents/kg                 |
| Alternative Solutions                                                                                                                                                                                        |
| Vet weight per \$ $\frac{2.5}{11.20} = 0.2232$<br><br>Supermarket weight per \$ $\frac{1.5}{6.25} = 0.240$<br><br>Supermarket better buy, more food /\$ ( 0.017 kg/\$)                                       |
| Specific behaviours                                                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>✓ Find Vet Cost per kg</li> <li>✓ Finds supermarket cost per kg</li> <li>✓ States that supermarket is cheaper by 31 cents/kg or better value by 0.017kg/\$</li> </ul> |

**See next page**

**Question 10**

**(8 marks)**

At the start of this year, an art supplies wholesaler in Australia needed to buy 3000 artist brushes and collected the following price information from overseas manufacturers:

| Country of manufacturer                              | Hong Kong | Canada | Thailand    |
|------------------------------------------------------|-----------|--------|-------------|
| Number of artist brushes per pack                    | 500       | 200    | 300         |
| Cost per pack (in local currency, includes shipping) | HK\$6850  | C\$460 | 17 000 baht |

At the time, the relevant exchange rates for one Australian dollar were 5.495 Hong Kong dollars, 0.898 Canadian dollars and 23.35 Thai baht.

- (a) Determine the total cost in Australian dollars of buying 3000 brushes from the Canadian manufacturer. **(3 marks)**

| <b>Solution</b>                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Will need $3000 \div 200 = 15$ packs.                                                                                                                                                           |
| Cost in Canadian dollars: $15 \times 460 = 6900$ .                                                                                                                                              |
| Cost in Australian dollars: $6900 \div 0.898 = \$7683.74$                                                                                                                                       |
| <b>Specific behaviours</b>                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>✓ indicates number of packs required</li> <li>✓ indicates total cost in local currency</li> <li>✓ calculates total cost in Australian dollars</li> </ul> |

- (b) Determine the cost per brush from the Thai manufacturer in Australian dollars. **(2 marks)**

| <b>Solution</b>                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------|
| Price in Australian dollars: $17\ 000 \div 23.35 = \$728.05$                                                                     |
| Unit price: $728.05 \div 300 = \$2.43$                                                                                           |
| <b>Specific behaviours</b>                                                                                                       |
| <ul style="list-style-type: none"> <li>✓ converts foreign price to Australian dollars</li> <li>✓ calculates unit cost</li> </ul> |

- (c) Determine, with justification, which of the three countries the wholesaler should use in order to minimise the unit cost per brush. **(3 marks)**

| <b>Solution</b>                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit price for Canadian manufacturer: $7683.74 \div 3000 = \$2.56$                                                                                                                                                          |
| Hong Kong pack price in Australian dollars: $6850 \div 5.495 = \$1246.59$                                                                                                                                                   |
| Hong Kong unit price: $1246.59 \div 500 = \$2.49$                                                                                                                                                                           |
| Hence unit cost per brush from HK, C, T is \$2.49, \$2.56, \$2.43 and so for minimum cost the wholesaler should choose the Thai manufacturer.                                                                               |
| <b>Specific behaviours</b>                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>✓ indicates unit price for Canadian manufacturer</li> <li>✓ indicates unit price for Hong Kong manufacturer</li> <li>✓ correctly identifies country for minimum unit cost</li> </ul> |

**Question 11****(8 marks)**

Information about the shares of three companies listed on the Australian Securities Exchange (ASX) towards the end of last year is shown in the table below.

| ASX Share Code | Market value of share (\$) | Earnings per share (\$) | Dividend per share (\$) | Percentage dividend |
|----------------|----------------------------|-------------------------|-------------------------|---------------------|
| NST            | 9.47                       | 1.046                   | 0.18                    | <b>A</b>            |
| RHC            | 70.45                      | 1.804                   | 1.55                    | 2.20%               |
| GMG            | 26.09                      | 1.221                   | <b>B</b>                | 1.15%               |

- (a) Determine the value of **A** and the value of **B** in the table. (2 marks)

| Solution                         |
|----------------------------------|
| $A = 0.18 \div 9.47 = 1.90\%$    |
| $B = 26.09 \times 1.15\% = 0.30$ |
| Specific behaviours              |
| ✓ value of <b>A</b>              |
| ✓ value of <b>B</b>              |

- (b) Calculate the price-to-earnings ratio for each share and hence rank the stocks in order from lowest to highest price relative to earnings. (3 marks)

| Solution                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P/E ratios:<br>NST: $9.47 \div 1.046 = 9.05$<br>RHC: $70.45 \div 1.804 = 39.05$<br>GMG: $26.09 \div 1.221 = 21.37$<br><br>Ranking from lowest to highest is NST, GMG, RHC.<br><br><i>(Accuracy: Accept ratios that round to 9, 39 and 21)</i> |
| Specific behaviours                                                                                                                                                                                                                           |
| ✓ correctly calculates at least one P/E ratio<br>✓ all ratios correct<br>✓ correctly ranks in order from lowest to highest                                                                                                                    |

- (c) Calculate, to the nearest dollar, the total dividend paid to an investor who owned 915 shares in NST, 230 shares in RHC and 412 shares in GMG. (3 marks)

| Solution                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Individual stock dividends:<br>NST: $915 \times 0.18 = \$164.70$ or $9.47 \times 915 \times 0.019 = \$164.64$<br>RHC: $230 \times 1.55 = \$356.50$ or $70.45 \times 230 \times 0.022 = \$356.48$<br>GMG: $412 \times 0.30 = \$123.60$ or $26.09 \times 412 \times 0.0115 = \$123.61$<br><br>Totals: \$644.80 or \$644.73. Hence total dividend is \$645 to nearest dollar. |
| Specific behaviours                                                                                                                                                                                                                                                                                                                                                        |
| ✓ correctly calculates dividend for at least one stock<br>✓ all stock dividends correct (using either method)<br>✓ correct total dividend                                                                                                                                                                                                                                  |

Question 12

(9 marks)

Ash lives at home with her parents. Her take-home income and all the expenses used in her personal budget from the previous year are shown below, together with how often the amounts are incurred and updates that Ash needs to make for the current year.

Use 1 year = 52 weeks and 1 year = 26 fortnight throughout this question.

| Budget item                | Previous amount (\$) | Frequency   | Update             |
|----------------------------|----------------------|-------------|--------------------|
| Take-home income           | 2100.00              | Fortnightly | Increase by 3.5%   |
| Gym membership             | 22.00                | Weekly      | Increase by \$3.50 |
| Household contribution     | 140.00               | Weekly      | Increase by \$30   |
| Mobile phone               | 45.25                | Monthly     | No change          |
| Car running costs          | 150.00               | Weekly      | Decrease by 10%    |
| Car loan repayment         | 506.75               | Monthly     | No change          |
| Entertainment & eating out | 150.00               | Weekly      | Increase by 5%     |
| Other                      | 450.00               | Monthly     | Decrease by 5%     |

- (a) Determine the amount of her fortnightly take-home income for the current year, after the change indicated in the update column. (2 marks)

| Solution                                       |
|------------------------------------------------|
| $I = 1.035 \times 2100$ $= \$2173.50$          |
| Specific behaviours                            |
| ✓ indicates correct method<br>✓ correct amount |

- (b) Use the information in the update column to complete the missing entries in the table below for Ash's budget for the whole of this current year.

Use 1 year = 52 weeks and 1 year = 26 fortnights

(3 marks)

| Budget item                | Amount this year (\$) | Frequency   |
|----------------------------|-----------------------|-------------|
| Take-home income           | <b>2173.50</b>        | Fortnightly |
| Gym membership             | <b>25.50</b>          | Weekly      |
| Household contribution     | <b>170.00</b>         | Weekly      |
| Mobile phone               | 45.25                 | Monthly     |
| Car running costs          | <b>135.00</b>         | Weekly      |
| Car loan repayment         | 506.75                | Monthly     |
| Entertainment & eating out | <b>157.50</b>         | Weekly      |
| Other                      | 427.50                | Monthly     |

| Solution                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| $\$2173.50, \quad 22.00 + 3.50 = \$25.50, \quad 140.00 + 30.00 = \$170.00$ $150.00 \times 0.90 = \$135.00, \quad 150 \times 1.05 = \$157.50$ |
| Specific behaviours                                                                                                                          |
| ✓ first three entries (income and flat increases) correct<br>✓ fourth entry (% decrease) correct                                             |

After subtracting her expenses from her take-home income, Ash calculated that at the end of last year she was left with a surplus of \$18 552 to use for a holiday or deposit in a savings account.

- (c) Ash hopes that her budget updates for the current year will lead to an increase in her annual surplus. Determine, with justification, whether this is the case. (4 marks)

| Solution                                                                                                                                                                                                                                                             |                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Income:                                                                                                                                                                                                                                                              | $I = 2173.50 \times 26 = \$56\,511$                                                  |
| Weekly costs:                                                                                                                                                                                                                                                        | $C_W = (25.50 + 170.00 + 135.00 + 157.50) \times 52$ $= 488 \times 52$ $= \$25\,376$ |
| Monthly costs:                                                                                                                                                                                                                                                       | $C_M = (45.25 + 506.75 + 427.50) \times 12$ $= 979.50 \times 12$ $= \$11\,754$       |
| Annual surplus:                                                                                                                                                                                                                                                      | $S = 56\,511 - (25\,376 + 11\,754)$ $= 56\,511 - 37\,130$ $= \$19\,381$              |
| Hence her annual surplus has slightly increased from \$18 552 to \$19 381 - which is as she had hoped.                                                                                                                                                               |                                                                                      |
| Specific behaviours                                                                                                                                                                                                                                                  |                                                                                      |
| <ul style="list-style-type: none"> <li>✓ correctly adjusts fortnightly income to annual</li> <li>✓ correctly adjusts weekly costs to annual</li> <li>✓ correctly adjusts monthly costs to annual</li> <li>✓ calculates surplus and notes it has increased</li> </ul> |                                                                                      |



**Question 13**

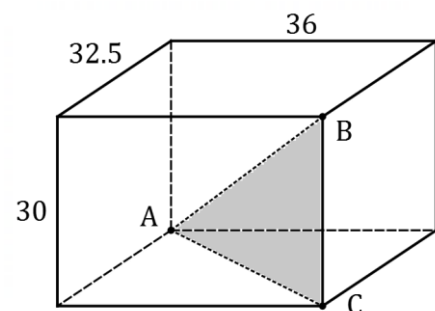
**(8 marks)**

Six decorative spheres are packed into a cuboid shaped box. Each sphere has a diameter of 14 cm, and the internal dimensions of the box are 36 cm by 32.5 cm by 30 cm.

- (a) Determine the percentage of the space inside the box that the spheres occupy. (4 marks)

| Solution                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volume of box: $V_B = 36 \times 32.5 \times 30 = 35\,100 \text{ cm}^3$ .                                                                                                     |
| Volume of one sphere of radius 7 cm: $V_S = \frac{4}{3} \times \pi \times 7^3 = 1436.76 \text{ cm}^3$ .                                                                      |
| Volume occupied by 6 spheres: $V_6 = 1436.76 \times 6 = 8621 \text{ cm}^3$ .                                                                                                 |
| Percentage of space occupied: $8621 \div 35\,100 = 24.6\%$                                                                                                                   |
| Specific behaviours                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>✓ volume of box</li> <li>✓ volume of one sphere</li> <li>✓ volume occupied by all spheres</li> <li>✓ calculates percentage</li> </ul> |

- (b) A thin triangular piece of card  $ABC$  is inserted into the box to limit the movement of the spheres, as shown in the diagram.



Determine the perimeter of the right triangle  $ABC$ .

**(4 marks)**

| Solution                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Length of diagonal of base: $AC = \sqrt{32.5^2 + 36^2} = 48.5 \text{ cm}$ .                                                                                                                                                                 |
| Length of diagonal of box: $AB = \sqrt{48.5^2 + 30^2} = 57.0 \text{ cm}$ .                                                                                                                                                                  |
| Hence perimeter of card is $48.5 + 57.0 + 30 = 135.5 \text{ cm}$ .                                                                                                                                                                          |
| Specific behaviours                                                                                                                                                                                                                         |
| <ul style="list-style-type: none"> <li>✓ indicates correct use of Pythagoras' Theorem</li> <li>✓ calculates length of diagonal of base</li> <li>✓ calculates length of diagonal of box</li> <li>✓ calculates length of perimeter</li> </ul> |

## Question 14

(7 marks)

The table below and matrix **N** show the number of trips (in thousands) made in 2006 by visitors to three Australian cities categorised by reason (VF;visiting friends or Holiday).

| (000's)  | VFR  | Holiday |
|----------|------|---------|
| Perth    | 1125 | 1060    |
| Brisbane | 1827 | 1544    |
| Darwin   | 91   | 313     |

$$\mathbf{R} = \begin{bmatrix} 1.02 & 0 & 0 \\ 0 & 1.045 & 0 \\ 0 & 0 & 1.03 \end{bmatrix}, \quad \mathbf{N} = \begin{bmatrix} 1125 & 1060 \\ 1827 & 1544 \\ 91 & 313 \end{bmatrix}$$

Visitor numbers (in both categories) to Perth, Brisbane and Darwin increased by 2%, 4.5% and 3% per year respectively over the following years and this information is shown in matrix **R**.

- (a) Determine matrix  $\mathbf{R} \times \mathbf{N}$ , rounding entries to the nearest whole number. (2 marks)

| Solution                                                                                                           |                                                                        |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| $\mathbf{R} \times \mathbf{N} =$                                                                                   | $\begin{bmatrix} 1148 & 1081 \\ 1909 & 1613 \\ 94 & 322 \end{bmatrix}$ |
| Specific behaviours                                                                                                |                                                                        |
| <ul style="list-style-type: none"> <li>✓ at least one row is correct</li> <li>✓ correct matrix, rounded</li> </ul> |                                                                        |

- (b) How many more VR trips were made to Brisbane in 2007 compared to the same type of trip to Perth in 2007? (2 marks)

| Solution                                                                                                                                            |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $1909 - 1148 = 761$<br>There were 761 000 more trips.                                                                                               |  |
| Specific behaviours                                                                                                                                 |  |
| <ul style="list-style-type: none"> <li>✓ identifies both required entries</li> <li>✓ states difference of trips, adjusting for thousands</li> </ul> |  |

- (c) The matrix  $\mathbf{R}^n \times \mathbf{N}$  will show the number of trips (in thousands) by category in the  $n^{\text{th}}$  year after 2006. Determine, with justification, the year in which the number of holiday trips to Brisbane first exceeded 2 000 000. (3 marks)

| Solution                                                                                                                                                                              |                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| $\mathbf{R}^6 \times \mathbf{N} =$                                                                                                                                                    | $\begin{bmatrix} 1267 & 1194 \\ 2379 & \mathbf{2011} \\ 109 & 374 \end{bmatrix}$ . Hence after 6 years, in 2012. |
| Specific behaviours                                                                                                                                                                   |                                                                                                                  |
| <ul style="list-style-type: none"> <li>✓ indicates value of <math>n</math></li> <li>✓ states correct year</li> <li>✓ justification (shows matrix, element or other method)</li> </ul> |                                                                                                                  |

**Question 15**

**(8 marks)**

The image shows some recently harvested grain stored in a conical pile on level ground.



The top of the pile is 12 m above the ground and the radius of the base is 35 m.

- (a) Determine the volume of grain contained in the pile.

**(2 marks)**

| Solution                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------|
| $V = \frac{1}{3} \times \pi \times 35^2 \times 12$ $= 15\,394 \text{ m}^3$                                                                      |
| Specific behaviours                                                                                                                             |
| <ul style="list-style-type: none"> <li>✓ shows correct use of volume formula</li> <li>✓ calculates volume (correct to at least 3 sf)</li> </ul> |

- (b) Trucks that can carry a load of 28 tonnes will be used to transport the grain to the local port. If one cubic metre of grain weighs 630 kg, determine the number of truck loads of grain required to move the entire pile of grain to the port. Note that there are 1000 kg in one tonne.

**(2 marks)**

| Solution                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------|
| $W = 15\,394 \times 630 \div 1000$ $= 9698 \text{ tonnes}$<br>$9698 \div 28 = 346.4$ <p>Hence will need 347 truck loads.</p>    |
| Specific behaviours                                                                                                             |
| <ul style="list-style-type: none"> <li>✓ calculates weight of grain in tonnes</li> <li>✓ calculates number of trucks</li> </ul> |

- (c) Determine the slant height of the conical pile.

**(2 marks)**

| Solution                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------|
| $s = \sqrt{12^2 + 35^2}$ $= \sqrt{1369} = 37 \text{ m}$                                                                                 |
| Specific behaviours                                                                                                                     |
| <ul style="list-style-type: none"> <li>✓ shows correct use of Pythagoras' Theorem</li> <li>✓ calculates correct slant height</li> </ul> |

- (d) When rain is forecast, the sloping surface of the grain can be covered with tarpaulins. Calculate the area of grain that needs to be covered.

**(2 marks)**

| Solution                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------|
| $A = \pi \times 35 \times 37$ $= 4068 \text{ m}^2$                                                                       |
| Specific behaviours                                                                                                      |
| <ul style="list-style-type: none"> <li>✓ uses correct part of surface area formula</li> <li>✓ calculates area</li> </ul> |

## Question 16

(10 marks)

- (a) Kelly takes a short-term loan of \$2400 at 9.6% per annum simple interest for 50 days. Determine the sum of the principal and interest that Kelly must repay at the end of the loan period. (3 marks)

| Solution                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $I = \frac{2400 \times 9.6 \times 50}{100 \times 365} = 31.56$ <p>Repay <math>2400 + 31.56 = \\$2431.56</math>.</p>                                                          |
| Specific behaviours                                                                                                                                                          |
| <ul style="list-style-type: none"> <li>✓ correct expression to calculate simple interest</li> <li>✓ calculates simple interest</li> <li>✓ calculates sum to repay</li> </ul> |

- (b) An amount of \$15 000 is invested at 4.5% per annum compounded annually for 3 years. Determine the total interest earned on this investment over the 3 years. (3 marks)

| Solution                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $A = 15\,000(1.045)^3 = 17\,117.49$ <p>Interest earned is <math>17\,117.49 - 15\,000 = \\$2117.49</math>.</p>                                                             |
| Specific behaviours                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>✓ correct expression to calculate future value</li> <li>✓ calculates future value</li> <li>✓ calculates interest earned</li> </ul> |

- (c) At the start of last year, Syed was employed as a manager with a salary of \$86 000. During the year, Syed was awarded a 6.5% pay rise followed by another rise of 7.7% at the start of this year.

- (i) Determine Syed's salary at the start of this year. (2 marks)

| Solution                                                                                                      |
|---------------------------------------------------------------------------------------------------------------|
| $S = 86\,000 \times 1.065 \times 1.077 = \$98\,642.43$                                                        |
| Specific behaviours                                                                                           |
| <ul style="list-style-type: none"> <li>✓ indicates correct method</li> <li>✓ calculates new salary</li> </ul> |

- (ii) Determine the percentage increase in Syed's salary from the start of last year to the start of this year. (2 marks)

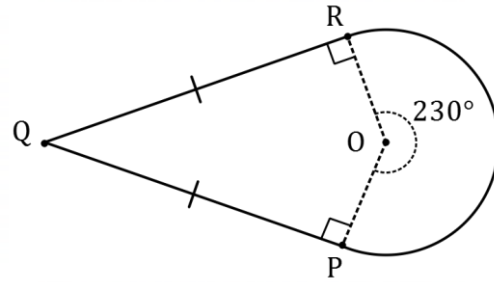
| Solution                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{98\,642.43 - 86\,000}{86\,000} = \frac{12\,642.43}{86\,000} = 14.7\% \text{ increase.}$ <p>or</p> $1.065 \times 1.077 = 1.147005 \rightarrow 14.7\% \text{ increase.}$ |
| Specific behaviours                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>✓ indicates correct method</li> <li>✓ correct percentage increase</li> </ul>                                                           |

**Question 17**

**(8 marks)**

The diagram (not to scale) shows a fabric panel used in the construction of a kite.

The panel consists of a  $230^\circ$  sector of a circle with centre  $O$  and radius 50 cm, and quadrilateral  $OPQR$  in which  $QR = QP = 108$  cm.



- (a) Calculate the area of the sector.

**(2 marks)**

| Solution                                                                                                      |
|---------------------------------------------------------------------------------------------------------------|
| $A = \frac{230}{360} \times \pi \times 50^2 = 5018 \text{ cm}^2$                                              |
| Specific behaviours                                                                                           |
| <ul style="list-style-type: none"> <li>✓ uses appropriate method</li> <li>✓ calculates sector area</li> </ul> |

- (b) Hence determine the total area of the fabric panel.

**(3 marks)**

| Solution                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Quadrilateral can be split into two right triangles.<br/> Area of a triangle:</p> $A = \frac{1}{2} \times 50 \times 108 = 2700 \text{ cm}^2$ <p>Hence total area is sector plus two triangles:</p> $A = 5018 + 2 \times 2700 = 10\,418 \text{ cm}^2$ |
| Specific behaviours                                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>✓ splits quadrilateral into two right-triangles</li> <li>✓ area of one right-triangle</li> <li>✓ calculates total area</li> </ul>                                                                                |

- (c) The panel is to be edged with tape. Calculate the length of tape required.

**(3 marks)**

| Solution                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Perimeter of sector, <math>P</math>:</p> $P = \frac{230}{360} \times 2 \times \pi \times 50 = 201 \text{ cm}$ <p>Total length of tape, <math>L</math>:</p> $L = 201 + 2 \times 108 = 417 \text{ cm}$ |
| Specific behaviours                                                                                                                                                                                     |
| <ul style="list-style-type: none"> <li>✓ indicates correct method for sector perimeter</li> <li>✓ calculates sector perimeter</li> <li>✓ calculates total length of tape</li> </ul>                     |

**Question 18****(9 marks)**

Covid created a travel bubble in 2021 where many people from both Australia and New Zealand would travel only across the Tasman.

Mary changed some Australian dollars (\$AUD) into New Zealand dollars(\$NZD).

The exchange rate was given as \$1AUD = \$1.07NZD.

She exchanged \$2000 AUD.

- (a) How many \$NZD did she receive?

**(1 marks)**

| Solution                              |
|---------------------------------------|
| $2000 \times 1.07 = \$\text{NZD}2140$ |
| Specific behaviours                   |
| ✓ correct \$NZD currency given        |

- (b) After spending most of her \$NZD, she had \$300NZD left.  
What was the value of this in \$AUD?

**(2 marks)**

| Solution                                |
|-----------------------------------------|
| $\frac{300}{1.07} = \$\text{AUD}280.37$ |
| Specific behaviours                     |
| ✓ Divides by 1.07                       |
| ✓ Correct answer r/w                    |

- (c) Determine the conversion rate of \$NZD to \$1AUD.

**(2 marks)**

| Solution                        |
|---------------------------------|
| Rate = $\frac{1}{1.07} = 0.935$ |
| Specific behaviours             |
| ✓ divides 1 by 1.07             |
| ✓ Correct answer r/w            |

Mary's child had three notes; a \$50, a \$20 and a \$10, but she didn't know which were NZ and which were Australian

**(2 marks)**

- (d) Using all three notes, what was the **maximum** value possible in Australian dollars, if one of the notes was from NZ?

**(2 marks)**

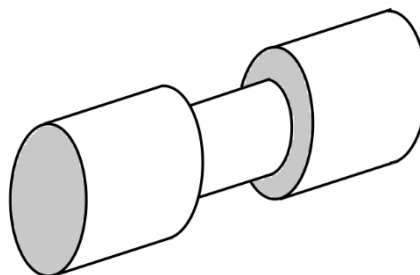
| Solution                                                    |
|-------------------------------------------------------------|
| Max value = $\frac{10}{1.07} + 20 + 50 = \$\text{AUD}79.35$ |
| Specific behaviours                                         |
| ✓ shows 10 divided by 1.07 (ZN note)                        |
| ✓ Correct max value                                         |

- (e) Using all three notes, what was the **minimum** value possible in New Zealand dollars, if one of the notes was from Australia? (2 marks)

| Solution                                                                                                         |
|------------------------------------------------------------------------------------------------------------------|
| Min value = $10 \times 1.07 + 50 + 20 = \$\text{NZD}80.70$                                                       |
| Specific behaviours                                                                                              |
| <ul style="list-style-type: none"><li>✓ Shows multiply by 1.07 (Aust note)</li><li>✓ Correct Min value</li></ul> |

**Question 19****(9 marks)**

A woodturner takes a solid cylindrical wooden pole of length 66 cm and radius 8 cm and uses a machine to reduce the radius of the middle third of the pole by 30%, as shown in the diagram (not to scale).



- (a) Calculate, with working, the volume of wood removed by the machine.

**(4 marks)**

| Solution                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Volume removed will be difference in volume of middle section of pole as a cylinder with height of <math>66 \div 3 = 22</math> cm and initial radius of 8 cm and final radius of <math>8 \times 0.7 = 5.6</math> cm.</p> $V_1 = \pi \times 8^2 \times 22 = 4423 \text{ cm}^3$ $V_2 = \pi \times 5.6^2 \times 22 = 2167 \text{ cm}^3$ <p>Hence volume removed is <math>4423 - 2167 = 2256 \text{ cm}^3</math>.</p> |
| Specific behaviours                                                                                                                                                                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"> <li>✓ explanation of method or describes each step</li> <li>✓ uses correct dimensions for one cylinder</li> <li>✓ correctly calculates volume of one cylinder</li> <li>✓ correct volume removed</li> </ul>                                                                                                                                                                        |



- (b) Explaining your method, determine the total surface area of the remaining object. (5 marks)

| Solution                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>TSA of one large end cylinder:</p> $A_1 = 2\pi \times 8^2 + 2\pi \times 8 \times 22$ $= 402.1 + 1105.84$ $= 1508 \text{ cm}^2$ <p>Area of curved surface of small middle cylinder:</p> $A_2 = 2\pi \times 5.6 \times 22 = 774 \text{ cm}^2$ <p>Area of one circular end of small middle cylinder:</p> $A_3 = \pi \times 5.6^2 = 98.5 \text{ cm}^2$ <p>TSA of remaining object:</p> $A = 2A_1 + A_2 - 2A_3$ $= 3016 + 774 - 197$ $= 3593 \text{ cm}^2$ |
| Specific behaviours                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>✓ explanation of method or describes each step</li> <li>✓ correct use of TSA formula for one cylinder</li> <li>✓ correctly calculates TSA of one cylinder</li> <li>✓ correct use of circle formula for cylinder ends</li> <li>✓ correct TSA of object</li> </ul>                                                                                                                                                  |

Supplementary page

Question number: \_\_\_\_\_

Supplementary page

Question number: \_\_\_\_\_

